The Physics and Engineering of Jet Engines

An Introduction to Air-Breathing Propulsion Systems

7.1 The Principle of Jet Propulsion

Every jet engine, regardless of its size or configuration, operates on a single unifying physical principle: Newton's third law of motion. An engine accelerates a mass of air (and, in the case of combustion engines, the products of burned fuel) rearward through the engine, and in reaction the engine itself is pushed forward. The net forward force produced by this exchange of momentum is called **thrust**. Unlike a propeller-driven aircraft, which relies primarily on an external airfoil to generate a pressure differential, a jet engine generates thrust entirely by processing air internally: drawing it in, compressing it, energizing it, and expelling it at high velocity.

The fundamental thrust equation, derived from the conservation of linear momentum applied to a control volume surrounding the engine, is written as:

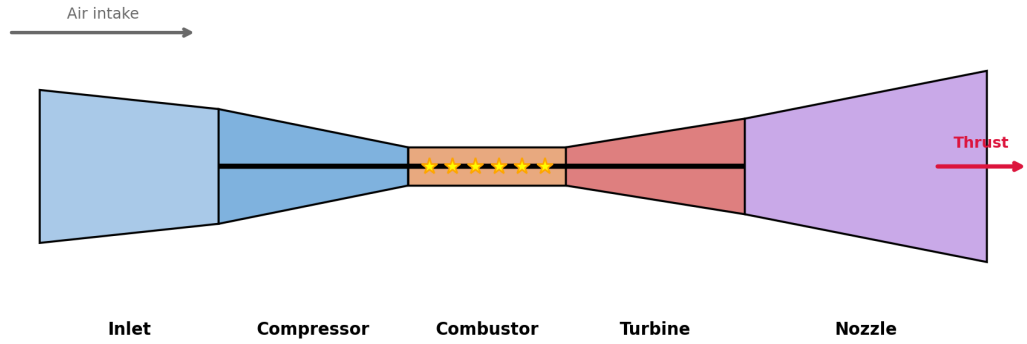
$$F = \dot{m}_a (V_e - V_0) + (p_e - p_0) A_e$$

where F is net thrust, $\dot{m}_a$ is the mass flow rate of air through the engine, V_e and V_0 are the exhaust and freestream (flight) velocities, p_e and p_0 are the exhaust and ambient static pressures, and A_e is the exhaust nozzle exit area. When the nozzle is *perfectly expanded* — exhaust pressure equal to ambient — the pressure term vanishes, giving the familiar momentum form $F = \dot{m}_a (V_e - V_0)$.

7.2 Anatomy of a Turbojet Engine

Figure 7.1 illustrates the five principal sections found in the simplest gas-turbine jet engine, the turbojet. Air enters through the **inlet**, which slows and diffuses the incoming flow before it reaches the **compressor**. The compressor consists of many rows of rotating and stationary airfoils (rotor and stator blades) that progressively raise the pressure of the air, typically by a factor of 15 to 40 in a modern engine. This high-pressure air then enters the **combustor**, where fuel is injected and burned continuously, raising the gas temperature to between 1800 K and 2000 K. The resulting high-energy gas expands through the **turbine**, which extracts just enough mechanical work to drive the compressor via a common shaft, before the remaining energy is converted to high-velocity exhaust in the **nozzle**, producing thrust.

Figure 1.1 — Schematic Cross-Section of a Turbojet Engine



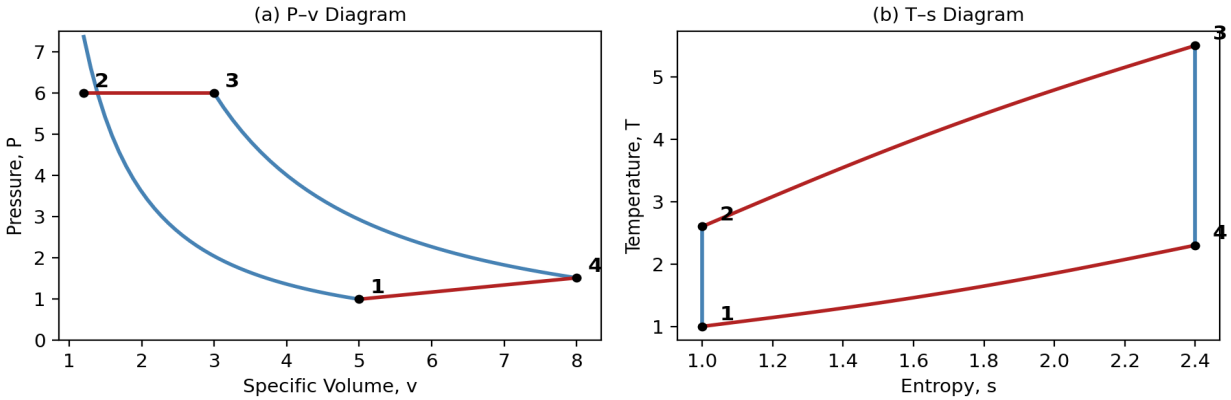
Air is drawn in at the inlet, compressed, mixed with fuel and burned in the combustor, then expanded through the turbine (which powers the compressor) before accelerating out the nozzle to generate thrust.

7.3 Thermodynamics: The Brayton Cycle

The thermodynamic behavior of every gas-turbine engine is idealized by the **Brayton cycle** (also called the Joule cycle), consisting of four processes performed on air treated as an ideal gas. Figure 7.2 shows this cycle on pressure–volume (P–v) and temperature–entropy (T–s) coordinates.

- **1 → 2, Isentropic compression:** Air is compressed adiabatically and reversibly in the compressor, raising both its pressure and temperature while entropy remains constant.
- **2 → 3, Constant-pressure heat addition:** Fuel is burned in the combustor, adding heat to the air at (ideally) constant pressure and sharply increasing its temperature.
- **3 → 4, Isentropic expansion:** The hot gas expands through the turbine and nozzle, converting thermal and pressure energy into mechanical work and kinetic energy.
- **4 → 1, Constant-pressure heat rejection:** In the idealized closed cycle, the exhausted gas is imagined to reject heat to the atmosphere at constant pressure, returning it to its initial state.

Figure 2.1 — The Ideal Brayton Cycle (1→2 Compression, 2→3 Heat Addition, 3→4 Expansion, 4→1 Heat Rejection)



The ideal Brayton cycle plotted in P–v and T–s coordinates. The enclosed area in the P–v diagram represents net specific work output; the enclosed area in the T–s diagram represents net heat converted to work.

The thermal efficiency of the ideal Brayton cycle depends only on the **pressure ratio**, $r_p = p_2/p_1$, and the ratio of specific heats, γ:

$$\eta_{th} = 1 - 1 / r_p^{(\gamma-1)/\gamma}$$

This remarkably simple result explains one of the central design trends in jet engine history: increasing the overall pressure ratio of the compressor is one of the most direct ways to improve fuel efficiency. Early turbojets of the 1940s operated with pressure ratios near 4:1; modern high-bypass turbofans exceed 40:1, and next-generation designs are pushing beyond 50:1.

7.4 Families of Air-Breathing Jet Engines

While all gas-turbine jet engines share the same Brayton-cycle core, engineers have developed several architectures optimized for different flight regimes.

Turbojet

The simplest configuration: all incoming air passes through the compressor, combustor, and turbine before exiting through a single nozzle. Turbojets produce high-velocity, low-mass-flow exhaust, which gives excellent performance at supersonic speeds but poor fuel efficiency at typical subsonic airliner cruise speeds.

Turbofan

A large fan is added ahead of (or integrated with) the compressor. Most of the air accelerated by the fan bypasses the engine core entirely, mixing with the core exhaust downstream or exiting through a separate cold nozzle. By accelerating a much larger mass of air to a lower velocity, turbofans achieve substantially higher propulsive efficiency, following directly from the identity that propulsive power for a given thrust is minimized when exhaust velocity is close to flight velocity. This makes the turbofan the dominant architecture for commercial aviation today.

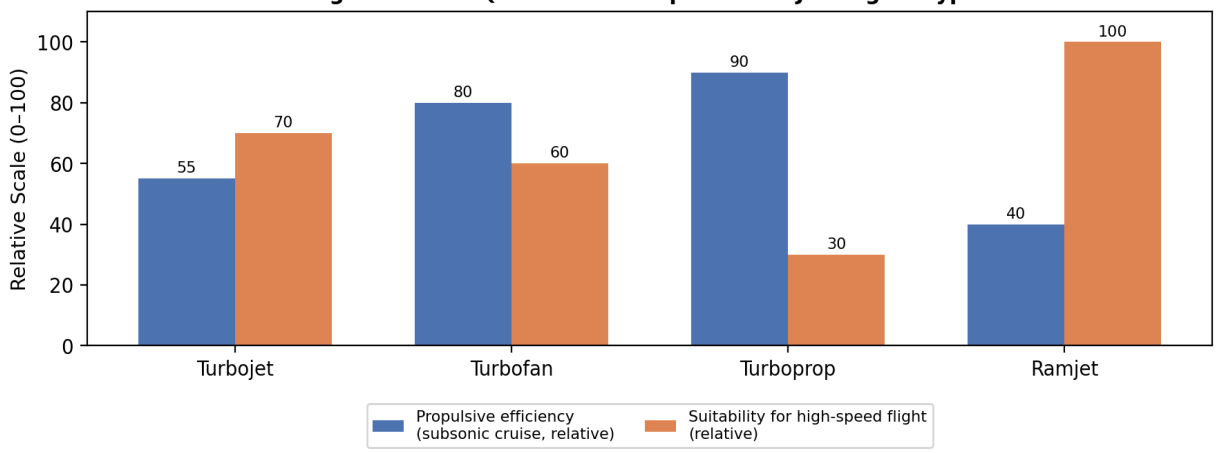
Turboprop and Turboshaft

Here, the turbine extracts nearly all of the gas energy to drive an external propeller (turboprop) or rotor (turboshaft) through a reduction gearbox, rather than accelerating exhaust gas for thrust. This yields the highest propulsive efficiency of any configuration at low subsonic speeds, but efficiency falls sharply as flight speed approaches the transonic regime because propeller blade tips encounter compressibility losses.

Ramjet

The ramjet eliminates rotating machinery altogether. At high supersonic speeds, the forward motion of the vehicle itself compresses the incoming air sufficiently for combustion, and expansion through a fixed nozzle produces thrust. Ramjets cannot produce static thrust and are only viable above roughly Mach 1, making them suited to missiles and high-speed research vehicles rather than general aviation.

Figure 4.1 — Qualitative Comparison of Jet Engine Types



A qualitative comparison of jet engine families. Turbofans dominate subsonic efficiency; turbojets and ramjets favor high-speed flight; turboprops excel at low speed but lose their advantage as flight speed increases. Values are illustrative, not measured data.

7.5 Worked Example

EXAMPLE 7.1 — Computing Net Thrust

A turbofan engine ingests air at a mass flow rate of 400 kg/s while cruising at a flight speed of 250 m/s. The engine accelerates this air to an average exhaust velocity of 320 m/s. The nozzle is perfectly expanded, so the pressure term in the thrust equation is zero. Find the net thrust produced.

Solution. Using the simplified thrust equation for a perfectly expanded nozzle:

$$F = \dot{m}_a (V_e - V_0)$$

F = (400 kg/s)(320 m/s – 250 m/s) = (400)(70) = 28,000 N ≈ 28.0 kN

This corresponds to roughly 6,300 lbf of thrust — a useful sanity check is that this figure sits in the range typical of a single engine on a modern narrow-body regional aircraft.

7.6 Key Concepts Summary

Term	Definition
Thrust (F)	Net forward force from momentum exchange with the working fluid
Pressure ratio (r_p)	Ratio of compressor exit to inlet stagnation pressure
Bypass ratio	Ratio of fan bypass mass flow to core mass flow in a turbofan
Propulsive efficiency	Fraction of kinetic energy imparted to the exhaust that converts to useful thrust power
Thermal efficiency	Fraction of fuel heat energy converted into useful shaft or jet work

7.7 End-of-Chapter Exercises

Exercise 7.1

An ideal Brayton cycle has a pressure ratio of 12. Taking $\gamma = 1.4$ for air, compute the thermal efficiency of the cycle.

Solution

$\eta_{th} = 1 - (1/r_p)^{(\gamma-1)/\gamma} = 1 - (1/12)^{0.2857}$

$(1/12)^{0.2857} \approx 0.535$, so $\eta_{th} \approx 1 - 0.535 =$ **0.465, or about 46.5%**.

Exercise 7.2

A turbojet flying at 200 m/s has an air mass flow rate of 50 kg/s and an exhaust velocity of 600 m/s. The nozzle is perfectly expanded. (a) Compute the net thrust. (b) Compute the propulsive power delivered to the aircraft (thrust × flight speed).

Solution

(a) $F = \dot{m}_a (V_e - V_0) = (50)(600 - 200) = (50)(400) =$ **20,000 N (20 kN)**.

(b) Propulsive power $= F \times V_0 = (20{,}000\ \text{N})(200\ \text{m/s}) =$ **4,000,000 W = 4.0 MW**.

Exercise 7.3

Explain, using the concept of propulsive efficiency, why a turbofan with a bypass ratio of 8:1 is generally more fuel-efficient at subsonic cruise than a pure turbojet, even though the turbofan produces a lower exhaust velocity.

Solution

Propulsive efficiency is highest when the exhaust velocity is only slightly greater than the flight velocity, because the kinetic energy left behind in the exhaust jet (the "wasted" energy) is proportional to the square of the velocity difference, while the thrust delivered is only proportional to the difference itself. A turbofan achieves the same net thrust as a turbojet by accelerating a much larger mass of bypass air through a smaller velocity increment, rather than accelerating a small mass of air through a large velocity increment. This produces the same momentum change (and hence the same thrust) while leaving far less residual kinetic energy in the exhaust, so a larger fraction of the engine's power output does useful work propelling the aircraft — that is, the propulsive efficiency is higher and less fuel energy is wasted as jet noise and heat.

Further Reading: Hill, P. & Peterson, C., *Mechanics and Thermodynamics of Propulsion*; Mattingly, J. D., *Elements of Gas Turbine Propulsion*.